

Universe24: one local integer rule on the cube's lattice, the click as its one non-local act, and the universe it builds without a free number

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September 2026

### Abstract

Can one local rule in whole numbers, on a cubic lattice read only through detectors' clicks, carry the quantum and the relativistic forms? The law is one line, Rule3: at every Node, each interval, a second-order step of a family's level from its two levels and its six neighbours' arrivals, with integer coefficients from the family's pair and the Node's paces, and one Euclidean division whose remainder stays at the Node. A quantum moves whole when a remainder crosses the quantum's action: the count's line, Rule3 read on its own conserved current, whose division is the click, the one non-local act. Exact on the lattice: reversibility bit for bit, a conserved quadratic form, the count's conservation, one quantum per click, Born's rule, no signalling through the click, and for a pair record the correlation  $\cos 2(a - b)$  with the Clauser–Horne–Shimony–Holt sum  $2\sqrt{2}$ . The families are bands of the line: three exact rotations of periods 6, 4 and 3, three tensor ranks from the three reads of the line, one free number per rank. A universe whose every number is the rule's is stated as a hypothesis: pairs from 1, 2 and 3, divisors 1, zero free numbers; there every bound body is one Node between the edge  $0.2255\Gamma$  and the horizon  $\Gamma \div 2$ , and the observer is a cluster of such Nodes. The forms of Einstein, Lorentz, de Broglie, Born, Malus and Bell are derived from the line as predictions; nothing is run and no agreement with nature is claimed.

**Keywords:** integer lattice model; one local rule; the click as the measurement; Bell correlations; discrete relativity; pre-registered simulation

## 1 Introduction

**The question.** The law of this paper is one line of integer arithmetic applied at every Node of a cubic lattice at every interval. Nothing else acts: no field equation beside it, no law of motion for a body, no rate, no draw, no real number. The paper states that line, proves what is exact on the lattice, derives what follows from the line alone, names what is assumed beside it, and writes down, before any run, what eighteen experiments read in clicks. The forms of Newton, Einstein, Lorentz, de Broglie, Born, Malus and Bell appear only on the comparison side, beside the law's own expectation; none is an input.

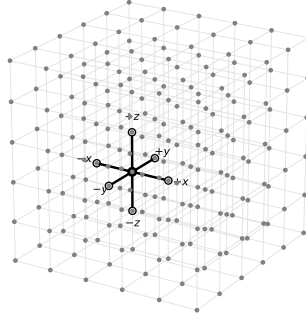
**The law in one line.** At every Node, each interval, for every family:

|                   |                                                                                                                                                                                                                                                                                                                                               |     |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Rule3:            | $w a_{\text{next}} + r' = \sum_{a \in \{x, y, z\}} R_a (\text{arr}_{+a} + \text{arr}_{-a}) + S a_{\text{now}} - w a_{\text{before}} + r, \quad 0 \leq r' < w,$ $w = 6 \text{ den } \Gamma^2, \quad R_a = 2 \text{ num } p_a^2,$ $S = 12 \text{ den } \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - 4 \text{ num } (p_x^2 + p_y^2 + p_z^2);$ | (1) |
| the count's line: | $T c_{\text{next}} + r' = T c_{\text{now}} + \sum_j F_{ij} + r, \quad 0 \leq r' < T,$ $F_{ij} = \text{weight} \times (\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j);$                                                                                                                                                        |     |
| the one write:    | a level at a Node gains $(w q(\text{Node}) + r) \text{ div } E$ , the remainder $r$ kept there.                                                                                                                                                                                                                                               |     |

In the first line  $a_{\text{now}}$ ,  $a_{\text{before}}$  and  $a_{\text{next}}$  are the family's level at the Node now, one interval before and one interval on;  $\text{arr}_{\pm a}$  is the arrival through the Port  $\pm a$ , the neighbour's level or its rotation by the Link's angle;  $r$  and  $r'$  are the remainder kept at the Node before and after the step;  $\Gamma$  is the Node clock, the pace of an empty Node;  $[\text{num}, \text{den}]$  is the family's pair, with  $\cos \omega_0 = \text{num}/\text{den}$  its rotation per interval at wave number zero;  $p_0$  is the Node's pace and  $p_a$  its pace along the axis  $a$ , both the Node clock lowered by the levels the family reads there; and  $\text{div}$  is the floor division. The second line is the same rule for the family of clicks:  $c$  is the count, the number of quanta at the Node;  $T$  is the quantum's action, the one unit of every family;  $F_{ij}$  is the current into the Node  $i$  from its neighbour  $j$  over the interval, the bilinear form of the record's two levels at the two ends; and the division is the click, one quantum moving whole exactly when the remainder crosses  $T$ . The third line is every write of a body or a record into a level:  $q(\text{Node})$  is the writer's own quantity there,  $w$  the one weight with which the writer's family reads the written family, and  $E$  the written row's divisor. Everything else is a declared number. Section 2 states the first line with its proofs, Section 3 the second and Section 4 the third; the paper cites the law's document [1] by the name of the section each formula stands in.

**The seven things, in the order they are built.** Each is named once, and the words are the law's [1, the objects].

- (1) *The GameBoard.* A box of Nodes in  $\mathbb{Z}^3$ , each joined to six neighbours by Links through its six Ports  $\pm x, \pm y, \pm z$ ; each axis periodic or open. The whole local information of a Node is its NodeState: the levels and remainders of the families there, and nothing else.
- (2) *The family and the record.* A family is one kind of physical value with its own declaration: its pair, its parts, its phase, what it reads and what it holds. Its record is a set of integer levels over the GameBoard, one per Node per level, with a remainder per Node; a family of two levels per Node holds a pair (re, im).
- (3) *Rule3.* The first line of Eq. (1), the one LocalRule: every family at every Node steps by it and by nothing else, a body's own record included.
- (4) *The click.* The second line: the family of clicks is a family like every other, its level the count, its wall  $T$ , its read the current; its division is the click, the law's one non-local act, at which a record ends whole where its last quantum leaves it.
- (5) *The one write.* The third line, in five instances: the hold, the source, the giving, the start and the recoil.
- (6) *The body.* A body is its count and its record: a set of Nodes on which a family's count stands, its record the standing wave the count binds, its momentum and its spin readings of that record's current. A body has no law of its own.
- (7) *The universe and the world.* Every physical number is a declaration outside the rule: the universe's, the families' rows and the universe's integers, and the world's, the bodies by their Nodes and counts, the detectors and the readings. The rule holds no number, no family's name and no exception.



one Node and its six Ports, the Links to its six neighbours

Fig. 1 The GameBoard in space: a cubic array of Nodes, here  $6 \times 6 \times 6$ , each joined to its six neighbours by Links through the Ports  $\pm x$ ,  $\pm y$ ,  $\pm z$  (the faint lines); one Node and its six neighbours marked, the Nodes one interval away from it, whose convex hull is the octahedron of Fig. 2. A Node keeps each family's two levels and its remainder there, and nothing else. Drawn from the definitions; no run.

**The tree: what needs what.** The paper's claims are sorted by what they need beyond the line, and each carries one of five marks: THEOREM, proved here for every GameBoard; DERIVED, from theorems by argument; COMPUTED, a number of the law's own computed before any run; ASSUMPTION, named where it is used; HYPOTHESIS, stated under its own name [1, the families from the rule, the tree]. The root is the line, Section 2. From it alone: the reversibility and the conserved form (Theorems 1 and 3), the guard's edge (Proposition 3), the count's conservation, one quantum per click, Born's rule and the pair's correlation  $\cos 2(a - b)$  (Theorems 4, 5, 6 and Proposition 4, Section 3), the three exact rotations and the three reads of the line with the ranks 1, 3 and 6 (Propositions 5 and 6, Section 6). With the assumption that the fields every family reads are the fields of unbounded reach: one free number per rank, DERIVED (Section 6.3). With the two named assumptions that the pairs are made of 1, 2 and 3 and that every divisor is 1: the rule's own universe, Section 7, in which the bound body is one Node with the horizon  $\Gamma \text{ div } 2$  (DERIVED) and the edge  $0.2255 \Gamma$  (COMPUTED), the cluster, the sieve and the wavelength 4 (DERIVED). With the assumption that every body is bound: the click that joins and parts, the observer as a cluster. Under hypotheses in their own names: the third, the conversion, the electron as the lightest band, the growing Node clock. The forms of nature, Section 8, are DERIVED from the line as predictions, and none is run. Not closed toward nature by name:  $\alpha$ ,  $m_e/m_p$ ,  $m_p/m_e$ , the quarks' charges, the generations and the band  $[0, 1]$  (Section 9).

**The reading rule and the notation.** Only a detector's click is a measurement; an experiment is compared with nature through its detectors' clicks alone [1, readings and measurements]. A number of the GameBoard's state (a level, a support, a total, a body's centre) is a diagnostic, labelled GAMEBOARD, never compared with nature and never pinned; a number computed from the algebra before any run is labelled COMPUTED; a declared integer of a world or a universe is a DECLARATION; a number of the bookkeeping outside the law is HOST. Every physics name in this paper (Born's, Malus's, Lorentz's, Tsirelson's) names the known form the law's result is set beside. Every symbol is named in English at its first use: a scalar plain, a vector in bold lowercase ( $\mathbf{n}$ ), a matrix or an operator in bold uppercase ( $\mathbf{M}$ ). `div` and `mod` are the floor division and its remainder, `isqrt` the integer square root.

**What this paper contributes, and what it does not claim.** New here: one explicitly stated integer rule whose reversibility, conserved form, count conservation, one-click theorem, Born's rule, no-signalling and pair correlation  $\cos 2(a - b)$  are theorems on the lattice; the families as bands of that rule, with one free number per tensor rank; a universe declaration with no free number, in which bound matter is one Node, and its consequences stated as algebra with their marks; and the forms of nature derived from the line as predictions, each beside the form it is compared with. Not claimed: any agreement with nature, since nothing is run here; a derivation of the Lorentz group, which is not a symmetry of the lattice (the lattice's symmetries are the 48 signed permutations of the axes); the fine-structure constant, the electron's mass ratios or the quarks' charges, which are open by name; or that nature is so. The paper is algebra: it holds no run, no code and no experiment, and the simulator that computes the rule is named only as the archive of the law's document.

## 2 The GameBoard and Rule3

This section states the object and proves what is exact on it: the lattice, the families, the one line with its coefficients, its reversibility, what its acts can and cannot form, the paces and their guard, the transport and the conserved form. Every formula is the law's [1, the GameBoard and the families; Rule3].

### 2.1 The objects

A *Node* is a physical location. The *GameBoard* is all the Nodes together, a box of shape  $= (N_x, N_y, N_z)$  Nodes; each axis is periodic (the box a torus on that axis) or open (a face beyond which no Node lies), as the world declares. Two Nodes that differ by one step along one axis are joined by a *Link*; each Node has six Links, one through each of its six *Ports*, named by their outward direction  $+x, -x, +y, -y, +z, -z$  (Fig. 1). An axis of one layer folds: its two Ports return the Node itself. The local information of a Node is its *NodeState*: the levels and remainders of the families at that Node, and nothing else. A change of a NodeState is an *Event*; the rule that makes it is a *LocalRule*, and there is one, Rule3.

A *family* is one kind of physical value with its own declaration. A family's *record* is a set of levels over the GameBoard, one integer per Node per level, with a remainder per Node; a family with two levels per Node holds a pair (re, im) at each Node. A *body* is a set of Nodes on which a family's *count* stands, with the values the body's writer declares (its content, its momentum, its spin, its moment); a body is one connected region.

**The universe's integers.** Every physical number is a declaration outside the rule: the universe's (the families and the universe's integers) and the world's (the bodies, the detectors, the readings); the rule holds no number of its own [1, a family's declaration]. The universe's integers are the Node clock  $\Gamma$ , the pace of an empty Node and every family's clock unit; the quantum's action  $T$ , the action of one quantum of any family, a unit as  $\Gamma$  is; the momentum unit  $Q_{\text{unit}}$ ; the twist table (Section 2.7); and the least number of remainder values a giving family's pair must leave at its giving Node, so that the residue of the ladder (Section 3.3) is spread over them. A family's row declares its pair [num, den], written as one integer  $m$  over the GameBoard's one denominator  $\Gamma$  (Section 6.2), its parts (1, 3 or 6 components, the tensor rank of what writes it), its phase (2 where it carries quanta, 1 where it never clicks), the families it reads, and where it holds a body's count, the row's divisor  $E_s$ . The weights of every read are 1, the sign is the body's and the action of one

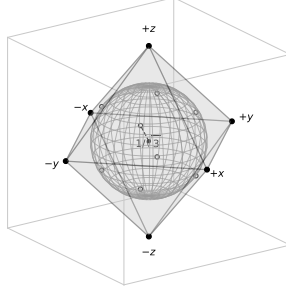


Fig. 2 The six Ports of one Node,  $\pm x, \pm y, \pm z$ , and the octahedron they span: the Nodes one interval away. The maps of the six Ports onto themselves that keep opposite Ports opposite are the 48 signed permutations of the axes, the symmetry group of the cube; the symmetric parts of a tensor under the 48 have the ranks 1, 3 and 6, the three ranks of Section 6. Drawn from the definitions; no run.

quantum is  $T$  for every family, so the declarations hold  $\Gamma, T$ , the matter pair and the two divisors, and nothing else of physics.

## 2.2 The line

At every Node, each interval, for every family, with the wall  $w$ , the read coefficient  $R_a$  on each axis  $a$ , the self coefficient  $S$ , the level  $a_{\text{now}}$  at the interval's start,  $a_{\text{before}}$  one interval earlier and the remainder  $r$  kept at the Node,

$$w a_{\text{next}} + r' = \sum_a R_a (\text{arr}_{+a} + \text{arr}_{-a}) + S a_{\text{now}} - w a_{\text{before}} + r, \quad 0 \leq r' < w, \quad (2)$$

$\text{arr}_{\pm a}$  the arrival through the Port  $\pm a$ : the neighbour's level, or its rotation by the Link's angle (Section 2.7); 0 beyond an open face; the Node's own level on a folded axis. The remainder is the Node's own and never travels. The coefficients at a Node, from its paces (Section 2.5) and its family's pair, are

$$w = 6 \text{ den } \Gamma^2, \quad R_a = 2 \text{ num } p_a^2, \quad S = 12 \text{ den } \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - 4 \text{ num } (p_x^2 + p_y^2 + p_z^2). \quad (3)$$

One floor division per Node per interval; the remainder stays at the Node. A held family's field (a real field of phase 1) steps at its row's pair with the pace 1 and the wall 3 den, at first order [1, the line].

**Proposition 1** (the vacuum's wave rule and the band). *At the vacuum's pace,  $p_0 = p_a = \Gamma$ , the self coefficient vanishes and Eq. (2) is, up to the remainder,  $a_{\text{next}} + a_{\text{before}} = \frac{\text{num}}{3 \text{ den}} S_6(a_{\text{now}})$  with  $S_6$  the sum over the six neighbours. A plane wave of wave number  $k$  along an axis rotates by  $\omega$  per interval with*

$$\cos \omega = \frac{\text{num}}{3 \text{ den}} (2 + \cos k), \quad \cos \omega_0 = \frac{\text{num}}{\text{den}} \text{ at } k = 0,$$

*and off an axis  $\cos k_x + \cos k_y + \cos k_z$  stands in place of  $2 + \cos k$ . The family's band of rotations is  $[\text{num}/(3 \text{ den}), \text{num}/\text{den}]$  in the cosine; light, the pair [1, 1], has  $\cos \omega = (2 + \cos k)/3$ , its group velocity at long wavelength  $1/\sqrt{3}$  Links per interval, and its band reaches  $\cos \omega = 1$ ; a pair with  $\text{den} > \text{num}$  has the gap  $\cos \omega_0 = \text{num}/\text{den}$  below 1, a rest rotation, a mass.*

*Proof.* With  $p_0 = p_a = \Gamma$ ,  $S = 12 \text{ den } \Gamma^2 - 12 \text{ den } \Gamma^2 + 12 \text{ num } \Gamma^2 - 12 \text{ num } \Gamma^2 = 0$  and  $R_a/w = \text{num}/(3 \text{ den})$ . For  $a_{\text{now}} = A \cos(kx - \omega t)$  along the axis  $x$ ,  $S_6(a_{\text{now}}) = (4 + 2 \cos k) a_{\text{now}}$  and  $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega a_{\text{now}}$ , so  $2 \cos \omega = \frac{\text{num}}{3 \text{ den}} (4 + 2 \cos k)$ .  $\square$

**Proposition 2** (the pace is conformal). *At a Node whose pace is  $p$  on every axis, every family's rotation at every wave number obeys*

$$\sin^2 \frac{\omega}{2} = \left(\frac{p}{\Gamma}\right)^2 \sin^2 \frac{\omega_{\text{vac}}(k)}{2},$$

$\omega_{\text{vac}}(k)$  the rotation of Proposition 1: the pace multiplies the clock of every wave and every bound mode alike.

*Proof.* At an isotropic pace  $S = 12 \text{ den } (\Gamma^2 - p^2)$  and  $R_a = 2 \text{ num } p^2$ , so  $2 \cos \omega = \frac{\text{num } p^2}{3 \text{ den } \Gamma^2} (4 + 2 \cos k) + 2 \left(1 - \frac{p^2}{\Gamma^2}\right) = 2 - 2 \left(\frac{p}{\Gamma}\right)^2 \left(1 - \frac{\text{num}}{3 \text{ den}} (2 + \cos k)\right) = 2 - 2 \left(\frac{p}{\Gamma}\right)^2 (1 - \cos \omega_{\text{vac}})$ , and  $1 - \cos \omega = 2 \sin^2(\omega/2)$ .  $\square$

Two clocks at one level shift alike,  $d \ln \omega / dc = 2 \tan(\omega/2) / (\omega p)$  per unit of the held level  $c$ , and a packet's fall in a pace gradient does not depend on its family or its shape beyond its band's own [1, the line]: this is the paper's equivalence principle, a reading of the line and no postulate. The clock's slowing is this pace: a record at a Node of pace  $p$  rotates and moves as a record at  $\Gamma$  does, with its intervals  $p/\Gamma$  as long.

### 2.3 The direction

**Theorem 1** (the step is reversible bit for bit). *Let  $\sigma$  be the direction, +1 forward and  $-1$  backward,  $\text{div}$  and  $\text{mod}$  the floor division and its remainder in  $[0, w)$ , and  $\rho$  the remainder carried in. With*

$$u = \sigma \left( \sum_a R_a \text{arr}_a + S x - w y \right) + \rho, \quad z = \sigma(u \text{ div } w), \quad \rho' = u \text{ mod } w,$$

*forward  $(x, y, \rho) = (a_{\text{now}}, a_{\text{before}}, r)$  gives  $(z, \rho') = (a_{\text{next}}, r')$ , and backward  $(x, y, \rho) = (a_{\text{now}}, a_{\text{next}}, r')$  gives  $(z, \rho') = (a_{\text{before}}, r)$ .*

*Proof.* Forward,  $\sigma = 1$ , the line is the definition of  $\text{div}$  and  $\text{mod}$ . Backward,  $\sigma = -1$ : write  $Y$  for the right side's first two terms,  $\sum_a R_a \text{arr}_a + S a_{\text{now}}$ . Then  $u = w a_{\text{next}} - Y + r'$  and  $z = -\lfloor u/w \rfloor = \lceil (Y - w a_{\text{next}} - r')/w \rceil$ . From the forward line,  $Y - w a_{\text{next}} - r' = w a_{\text{before}} - r$  with  $0 \leq r < w$ , whose ceiling divided by  $w$  is  $a_{\text{before}}$  exactly; and  $\rho' = u - w \lfloor u/w \rfloor = w a_{\text{before}} - (Y - w a_{\text{next}} - r') = r$ . So the backward call returns  $(a_{\text{before}}, r)$  bit for bit, and the reversibility of every run needs no second function [1, the direction].  $\square$

The whole interval is a bijection but for the click: the places of the interval (Section 2.9) run backward in reverse order, each primitive by its own inverse, and the count's line by the same line with the current reversed (Section 3.2).

### 2.4 The four acts and what they can form

Rule3's line is one act; declared coefficients make it four [1, the four acts]: (a) *the read*, the line over the GameBoard with the six arrivals; (b) *the one-Node step*, the line with the neighbours' reads declared 0,  $a_{\text{next}} + a_{\text{before}} = (a/b) a_{\text{now}}$ , the rotation by the angle whose doubled cosine is  $a/b$  (Chebyshev's recurrence); with  $a/b = 1$  and the coefficient on  $a_{\text{before}}$  declared 0 it keeps a level, and with  $a/b = 2$  from  $(0, 1)$  it counts,  $a_t = t$ ; (c) *the division*, the line with  $R = S = 0$ ,  $w a_{\text{next}} + r' =$  the numerator  $+ r$ , Euclid's division with the remainder kept; (d) *the load*, a declared integer in the numerator, how a count enters a level. A composition of the acts is a finite sequence of them, each with its coefficients (a pair, a weight, a wall, a load) declared for the world, applied to declared levels.

**Theorem 2** (what the acts can form). *Every composition of the four acts is a piecewise-linear function of the levels with integer slopes: a sum of linear forms and of floors of linear forms, composed. No such function is quadratic in the levels.*

*Proof.* Each act is a linear form of the levels or the floor of one; a floor of a linear form is piecewise linear with integer slopes; sums and compositions of piecewise-linear functions with integer slopes are of the same kind. The second difference of such a function in any one level is 0 almost everywhere, while the second difference of a product  $\text{now}_i \text{before}_j$  in the pair  $(\text{now}_i, \text{before}_j)$  is 1.  $\square$

What the theorem separates: every product of two levels in the law is a *booking*, a reading of a bilinear form with declared coefficients and no write (Section 3.1); the one thing written from a product is the count, by the count's line, and that product is Rule3's own conserved current (Theorem 3). *The three tests of every rule:* a line enters the law only if it is generic (one primitive with declared integers, no family's name and no kind), vector (one of the acts on the state, no root and no float) and local (its own record and the six neighbours, nothing kept at a Node beyond the law's numbers). A hypothesis that needs more is stated under its own name, outside the law.

## 2.5 The paces and the guard

The *pace* of a Node for a family is what its reads make of the Node clock  $[1, \text{the paces}]$ :

$$\begin{aligned} p_0 &= \Gamma - \sum_{\text{reads}} (\text{weight} \times \text{by} \times \text{the read family's time level}), \\ p_a &= p_0 - \sum_{\text{reads}} (\text{the same}) - \sum_{\text{reads}} (\text{weight} \times \text{by} \times \text{its } aa \text{ component div } 2), \quad a = x, y, z, \end{aligned} \quad (4)$$

one division per read per axis, rounded at the read: the pace is a coefficient of the interval and no level, so no remainder is kept for it. A positive weight is a hollow (the read level slows the clock, an attraction); a negative weight is a hill; by “ $q$ ” the weight is multiplied by the reading record's charge sign. A family with no reads steps at  $p_0 = p_a = \Gamma$ , the plain rule. The time level lowers the clock's pace  $p_0$  once and each axis's pace  $p_a$  once more, so a wave's speed along a Link falls by the level twice (the Link as well as the clock, as nature's light past a body bends by twice the clock's share) while a mode's rest rotation, from  $S$  alone, falls by the level once.

**Proposition 3** (the guard). *The step is stable at a Node exactly while  $0 < p \leq P$  with the edge*

$$P = \text{isqrt}(2 \text{den } \Gamma^2 \text{div } (\text{den} + \text{num})),$$

*$\Gamma$  for light and above  $\Gamma$  where  $\text{num} < \text{den}$ ; every pace at or below  $\Gamma$  is inside it.*

*Proof.* The mode at wave number  $\pi$  on every axis has  $S_6(a_{\text{now}}) = -6 a_{\text{now}}$ , so its factor is  $(S - 6R)/w = 2 - 2(1 + \text{num}/\text{den})(p/\Gamma)^2$ , which is  $-2 \text{num}/\text{den}$  at  $p = \Gamma$  and falls below  $-2$ , the edge of Chebyshev's recurrence, once  $p^2(\text{den} + \text{num})$  exceeds  $2 \text{den } \Gamma^2$ ; there the record grows without bound.  $P$  is the largest integer pace at which  $p^2(\text{den} + \text{num}) \leq 2 \text{den } \Gamma^2$  holds. Below 0 the pace is no clock.  $\square$

A hill lessens a hollow and never exceeds it; a pace outside the guard refuses the run, naming the Node. *The band at a pace*, a reading of the line and no new rule: at a Node of pace  $p$  on every axis a record's rotation at wave number  $k$  along an axis is

$$\cos \omega = 1 - \left(\frac{p}{\Gamma}\right)^2 \left(1 - \frac{\text{num}}{3 \text{den}} (2 + \cos k)\right), \quad (5)$$

so the band of rotations a body's Nodes carry narrows with  $p^2$ : a record whose rotation lies outside the band there is evanescent inside (for  $[1, 1]$ ,  $\cosh \kappa = (\Gamma/p)^2(1 - \cos k) - 1$  per Node) and is reflected whole, and one whose rotation lies inside is slowed to the band's group velocity. Light meets a total mirror exactly where  $p < \Gamma \sin(k/2)$ , that is where the held count  $c$  exceeds  $\Gamma(1 - \sin(k/2))$  ( $0.2929\Gamma$  at the wavelength 4 Links), and a window otherwise, with the step's partial share at each face. A body is a balance of a hollow and a hill of different ranges: a hollow alone in three dimensions binds no record (it collapses into its own well or disperses); gravity is the one-signed exception (Section 5).

## 2.6 The bound

The rule's total at a Node whose levels and arrivals stand at the amplitude  $A$  is at most  $6A|R| + A|S| + w(A + 1)$ . Every world declares integers under which this total sits below  $2^{63} - 1$ , the width of the integers, or the declaration is refused naming the total [1, the bound]. The amplitude unit  $A$  is derived, never written: the largest amplitude at which every total of an interval stays inside the width, the least of three, Rule3's  $2wA$  with  $w = 6\Gamma^3$  (Section 6.2), the count's line's  $6A^2$  and the transport's  $3d_1d_0(A + 1)$  with the largest denominator  $d$  of the twist table. With  $\Gamma = 10^4$  and  $A = 2^{20}$  Rule3's total is near  $3 \times 10^{18}$ , one third of the room (COMPUTED).

## 2.7 The transport

A family with a pair (re, im) at each Node and a read with a twist turns its arrivals by the Link's angle [1, the transport]. On the Port along the axis  $a$  in the sense  $\sigma$ , for each such read with the factor  $f$  (the weight times the record's own rotation for the twist "own", else the declared twist; by  $q$  the charge sign),

$$k = \sigma \times f \times (V_a \text{ at this Node} + V_a \text{ arrived through the Port}),$$

$V_a$  the read family's vector component along the axis; both ends form the same number with opposite signs, so the transport back is the inverse rotation. The universe's *twist table* gives, for the angle  $k$  in units of 1/unit radians, a Pythagorean triple  $(c, s, d)$  with  $c^2 + s^2 = d^2$  exactly:  $|k| = k_1F + k_0$  with  $F$  the fine table's length, the fine triple of  $k_0$  and the coarse triple of  $k_1$  composed,  $(c_1c_0 - s_1s_0, s_1c_0 + c_1s_0, d_1d_0)$ , with  $(c, -s, d)$  for  $k < 0$  and  $(1, 0, 1)$  at  $k = 0$ ; an angle beyond the table refuses the run naming the Port. The arriving pair is rotated to the nearest unit,

$$R_{\text{re}} = (2c \text{ re} - 2s \text{ im} + d) \text{ div } 2d, \quad R_{\text{im}} = (2s \text{ re} + 2c \text{ im} + d) \text{ div } 2d,$$

two read acts of Rule3 with the coefficients  $(2c, -2s)$  and  $(2s, 2c)$  on the two arrivals, the load  $d$ , the wall  $2d$ , the remainder not kept: a remainder carried across intervals is one to one only while  $d$  stands, and  $d$  changes with the angle every interval; the rounding is unbiased in the mean, and the inverse recomputes the same triple from the start's levels, exact. The record's own rotation is written once at the load from the record's pair (matter) or from its wavelength on light's dispersion (light), so no per-world number is declared; the charge's twist on a charged record is "own" by  $q$ , and no separate number exists.

**The second level and the self-source.** A phase-2 family runs the line on each of its two levels with the transported arrivals; with no twist the second level stays exactly zero when it starts zero. The self-source of a family with the unit  $P_2$  above 0,  $\Sigma_{\text{self}} = (\sum (a_j - a_i)^2) \text{ div } P_2$  over the six Links, is subtracted from the step's right side as  $w \Sigma_{\text{self}}$ : each difference a read act, its square a booking of the family's own levels, the division Rule3's;  $P_2 = 0$  turns the line off [1, the second level].

## 2.8 The conserved form

**Theorem 3** (the conserved form). *Let the paces stand. The read matrix  $2 \text{num } p_i^2$  is not symmetric where the paces differ, so the form carries the weight  $1/p_i^2$  on the Node terms and the plain current on the Links:*

$$E = \sum_i \frac{w (\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i}{p_i^2} - 2 \text{num} \sum_i \text{now}_i S_6(\text{before})_i.$$

$E$  is exact where the paces stand and changes by the work term where they move; with the rounding, each Node's remainder adds at most one unit of the level per interval to the walk of  $E$ . Its Link term, the plain current  $\text{num} (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$  through the Link  $ij$ , is the current of Rule3's own conservation law and the one product the click reads.

*Proof.* Without the rounding the line is  $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$  with  $\mathbf{M}$  the matrix whose diagonal is  $S_i/w$  and whose entries on the six Links are  $R/w$ . Let  $\mathbf{D}$  be the diagonal matrix of the weights  $1/p_i^2$ ; then  $\mathbf{DM}$  is symmetric, since  $R_a = 2 \text{num } p_a^2$  at the reading Node and the weight  $1/p_i^2$  cancels it on both sides of every Link. For a symmetric  $\mathbf{K} = \mathbf{DM}$  the quantity  $Q(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{D} \mathbf{x} + \mathbf{y} \cdot \mathbf{D} \mathbf{y} - \mathbf{x} \cdot \mathbf{K} \mathbf{y}$  satisfies  $Q(\mathbf{a}_{\text{next}}, \mathbf{a}_{\text{now}}) = Q(\mathbf{a}_{\text{now}}, \mathbf{a}_{\text{before}})$ : with  $\mathbf{a}_{\text{next}} = \mathbf{M} \mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}$ ,  $\mathbf{a}_{\text{next}} \cdot \mathbf{D} \mathbf{a}_{\text{next}} - \mathbf{a}_{\text{next}} \cdot \mathbf{K} \mathbf{a}_{\text{now}} = (\mathbf{M} \mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}) \cdot \mathbf{D} (\mathbf{M} \mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}) - (\mathbf{M} \mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}) \cdot \mathbf{D} \mathbf{M} \mathbf{a}_{\text{now}} = \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{a}_{\text{before}} - \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{M} \mathbf{a}_{\text{now}}$ , which is the other side. Multiplied by  $w$  this is  $E$  up to the Link term's sign convention [1, the conserved form].  $\square$

$E$  is a GameBoard reading. Its Node term at a Node,  $D_i = \text{now}_i^2 - \text{next}_i \times \text{before}_i$ , is the record's form there, and the record's quanta at the Node are  $D_i \text{div } T$ : the count is the record's form, and this is how a body's well is laid (Section 4).

## 2.9 The interval

The law steps every family from the state at the interval's start to the state at its end. Every read is of the start's values: the six neighbours' levels, the Node's own pace, the arrivals, never a value written in the same interval; one Link per interval and nothing in zero time. The interval has five places, in order [1, the interval]: (i) the clicking families' step, every component, with the transport through the Ports and the self-source; (ii) the bookings at the Ports (the current), the ladder, the takings and the givings, the count's line; (iii) the held families' step; (iv) the holds written (a body's values into its family's levels), the clicks' changes of a body's content, charge and momentum included; (v) the bodies on one Node: the feed, the induction, the spin's step, each a reading of the body's record and no write. The order within a place, where two primitives write one value, is declared once for every world. A click's writes enter at the next interval. Backward, the places run in reverse order and each primitive runs its own inverse.

## 3 The click

The click is the law's one non-local act and the only thing the Outside reads. This section states the booking, the count's line whose division is the click, the ladder that reads it today, and the theorems: the count is conserved, one quantum makes one click, Born's rule, and for a pair record the correlation  $\cos 2(a - b)$  with no signalling [1, the click].

### 3.1 The booking

The *current* of a record into Node  $i$  from its neighbour  $j$  through their Link over an interval is the bilinear form of its two levels at the two ends,

$$F_{ij} = \text{weight} \times (\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j), \quad (6)$$

the pair's second level added, the weight the family's common wall in the form's units; positive inward. A record of wave number  $k$  travelling from  $i$  to  $j$  has  $F_{ij} = -2 \times \text{weight} \times |\psi|^2 \sin \omega \sin k$ , below 0. It is antisymmetric,  $F_{ji} = -F_{ij}$ , and it is the Link term of the conserved form of Theorem 3: the current of Rule3's own conservation law. The booking reads it at a Port and writes nothing.

### 3.2 The count's line

The *family of clicks* is one family like every other: its level at a Node is the *count*  $c$ , the number of quanta there; its wall is  $T$ , the quantum's action; its read is the current. At every Node,

$$T c_{\text{next}} + r' = T c_{\text{now}} + \sum_j F_{ij} + r, \quad 0 \leq r' < T, \quad (7)$$

the sum over the six Ports of the current into the Node over the interval,  $r$  the remainder kept at the Node: Rule3 with the coefficient 1 on each axis's inflow through its two Ports,  $T$  on the count and the wall  $T$ . The *click* is the division: a quantum moves whole from one Node to the next exactly when the remainder crosses  $T$ , and never a fraction of one. The record ends where its last quantum leaves it; the count that arrives at a detector's Node is the detector's click, and a detector's count is the family of clicks' level at its Nodes; an emitter's giving is the count leaving through its outer Ports into the born record. The same line carries a body: the count follows its record's current at every Node, one quantum at a time, so a body moves without any accumulator of its position. The remainder at a Node starts at the ladder's origin,  $(2u + 1)T \text{ div } 2$  with  $u$  the record's residue (Section 3.3); a body's count is laid with its record, at every Node  $Tc + r$  the Node's share of the record's form plus the origin, so the count follows the norm with the margin  $T/2$  against the rounding's walk. The inverse is the same line with the current reversed, exact.

**Theorem 4** (the count is conserved). *Over any region of Nodes,  $\sum(Tc_i + r_i)$  changes in one interval only by the currents through the region's boundary Links.*

*Proof.* Summing Eq. (7) over the region, each interior Link's current enters one Node's right side as  $+F$  and its neighbour's as  $-F$  and cancels; what remains is the sum over the boundary Links, a telescoping sum, exact in integers. On a periodic GameBoard with no face the sum is constant to the bit.  $\square$

**The velocity.** The record's wave number is conserved by the record's line, and the count's line is the continuity equation of the current in integers, so the count's centroid moves at the current's velocity, with no leak beyond the ladder's remainder; a body at rest has zero net current at every Node in the mean and its count stays in place. A record of two levels with the phase  $k$  per Link carries its count at Rule3's group velocity,  $R \sin k / (w \sin \omega)$  (Section 5.4). Under the count's line a moving body is the moving mode of the generator, its count moved by its record's current.

### 3.3 The ladder

At a detector the click is read by the *ladder*, the same division act on a record's one-way inward flux [1, the ladder]. The data: the record's residue  $u$  (its remainder at its giving Node), its norm  $T$ , its

wheel  $V$ , hence its threshold  $\theta = (2u + 1)T/(2V)$ , fixed at the giving; the named detectors in their declared order with the face last. At every interval each Node  $i$  of a detector receives the one-way inward flux  $f_i(t) \geq 0$  of the record through its Ports (a Link between two Nodes of one detector is not a Port and carries no offer); the record's running total is  $C(t) = C(t - 1) + \sum_i f_i(t)$ . The click is at the first interval  $t^*$  with  $C(t^*) \geq \theta$ , at the Node whose segment of that interval's increment, laid in the ladder's order, contains  $\theta - C(t^* - 1)$ ; there the record ends, whole. The rungs are the division act,  $b_k = (2VC_k + \text{Total}) \text{ div } (2 \text{ Total})$ .

**Theorem 5** (one quantum, one click).  *$\theta$  is crossed exactly once.*

*Proof.*  $C$  is non-decreasing,  $\theta < T$ , and the total offered with the face last is at least  $T$  (what leaves offers everything at the face), so  $C$  crosses  $\theta$  once; the deletion of the record at once leaves nothing to cross it again.  $\square$

**Theorem 6** (Born's rule). *Let  $u$  be spread evenly over  $\{0, \dots, V - 1\}$ ; then the probability that the click lands at Node  $i$  is  $C_i(\infty)/T$ , the detector's share of the record's total inward flux, whatever the time profile.*

*Proof.*  $\theta$  is then spread evenly over  $V$  points in  $[0, T)$ . The segments  $[C(t - 1) + \sum_{j < i} f_j(t), C(t - 1) + \sum_{j \leq i} f_j(t))$ , one per (interval, Node), partition  $[0, C(\infty))$  and have the lengths  $f_i(t)$ ; so the probability of the click at interval  $t$  at Node  $i$  is  $f_i(t)/T$  to the grain  $1/V$ , and summing over  $t$  gives the share. The counts over many records are binomial about the shares, since the residues are equidistributed and not a permutation.  $\square$

The residue is the rule's own remainder at the giving Node, never declared and never drawn: chance lives in the remainders, which stay at the Nodes, bounded, never reset and never erased. A detector is one connected region of Nodes with one name; separate places are separate names. A detector *set* is one Node of the count's line with its Ports the set's outer Ports, the sum of its Nodes' lines with one remainder; Born's rule is the shares of the current among the set's Nodes.

### 3.4 The pair record and Bell's correlation

**The pair record, rank 2.** One record with two labels, its level at a Node the  $2 \times 2$  matrix  $\mathbf{M}$  over the labels' two directions, written as the identity by a crystal's giving click (the two labels identical), stepped by Rule3 alone at every Node it reaches (the step, the transport and the paces as any record's, component by component, Rule3 linear), spreading from the crystal's Node through the Ports to both sides; a polariser turns the labels at its own Nodes by its angle through the transport's exact triple, the rows on the one side and the columns on the other, reading nothing beyond them [1, the ladder, the pair record]. The crystal is a body of matter with one key and one Node: its taking click takes the arriving quantum whole into its count, and in the same interval its giving click lowers its count by one and gives the pair record at  $T \text{ div } 2$  by the division act, each label one quantum of the record's own unit, the half; the theorem of one quantum per click holds per label in the label's unit, and the pair's two clicks are the arriving quantum, a half and a half, as nature's down-conversion.

**The labels' clicks.** Each label clicks alone at its own detector, by its own ladder (its own residue, the two spread at the crystal's Node), at the interval the label reaches the detector's Nodes, reading nothing but the level there. The detector of the first label at the setting  $a$  offers its two sets the rows of  $\mathbf{R}(-a)\mathbf{M}$ , each row's booking one half, and the taking click of the row  $i$  ends that label and not the pair: the record goes on as rank 1, its level at every Node the row  $i$  of the turned matrix

there (the click, the one non-local act, ends one label at once everywhere and leaves the other on the GameBoard), so the second label at the setting  $b$  offers  $\mathbf{R}(-b)$  applied to that row.

**Proposition 4** (the pair's shares). *With  $\mathbf{M}$  the identity at the giving, the first click lands on either of its side's two sets with the share  $1/2$  at every setting; the second click lands on the set of the same index with the share  $\cos^2(a - b)$  and on the other with  $\sin^2(a - b)$ , whichever label clicks first. So over all pairs*

$$E(a, b) = \cos^2(a - b) - \sin^2(a - b) = \cos 2(a - b),$$

*each side's marginal is  $1/2$  at every setting of the other side (no signalling), and the Clauser–Horne–Shimony–Holt sum [3] at the settings  $0, \pi/4$  for one side and  $\pi/8, 3\pi/8$  for the other is  $S = 2\sqrt{2} = 2.83$ , Tsirelson's bound [4], derived from the local law with the click's one non-local act alone.*

*Proof.* The rows of the rotation  $\mathbf{R}(-a)$  are two unit vectors, so each row's booking is one half of the label's norm: the first click's shares are  $1/2$  and  $1/2$ . After the click of the row  $i$  the record's level is that row, the unit vector at the angle  $-a + i\pi/2$ ; the second polariser applies  $\mathbf{R}(-b)$ , whose components on the two sets are  $\cos(a - b)$  and  $\pm\sin(a - b)$  up to the index's quarter turn, so the shares are their squares.  $E$  is the difference of the shares of the same and the other index. Summed over the first click's index at  $1/2$  each, the second side's marginal is  $\frac{1}{2}(\cos^2 + \sin^2) = \frac{1}{2}$ , whatever  $a$ ; the first side's is  $1/2$  by the first step, whatever  $b$ . The sum  $S = E(a, b) - E(a, b') + E(a', b) + E(a', b')$  at  $a = 0, a' = \pi/4, b = \pi/8, b' = 3\pi/8$  has  $E(0, \pi/8) = E(\pi/4, \pi/8) = E(\pi/4, 3\pi/8) = \cos(\pi/4)$  and  $E(0, 3\pi/8) = \cos(3\pi/4) = -\cos(\pi/4)$ , which enters with the minus sign, so  $S = 4\cos(\pi/4) = 2\sqrt{2}$ .  $\square$

The correlation travels in the record alone, by Rule3, one Link per interval; the click ends one label everywhere at once and carries no information, since each side's marginal does not depend on the other's setting: the law is local in its transport and non-local in the click only, as Bell's theorem requires of any account that reaches  $2\sqrt{2}$  [2]. The pairs are the host's coincidence reading, not law: the two clicks' intervals within a window declared beside the world, as Aspect's [5]; the law has no coincidence set and no joint click, since no uniform time of events exists on the GameBoard. With switching polarisers, each side's angle is the one in force at the label's passage, by the polariser's own clock, the count act on the body. A click on a record with a partner label is entanglement and a click on a record with none is a measurement, and Bell's  $S$  reads that and nothing else [1, the universe is bound].

## 4 The one write and the primitives

**A family's write is one act.** Every write into a family's level on the GameBoard is  $(w \times q(\text{Node}) + r) \div E$  at a Node each interval, at both levels of the writer's rotation (the second the read act once more, halved), the division act with the remainder  $r$  carried at that Node [1, the primitives]. Here  $q(\text{Node})$  is the writer's own quantity there: for a body, the count declared there times its tensor  $(1, n_a/W, n_a n_b/W^2)$  with  $\mathbf{n}$  its momentum and  $W$  its wall; for a record, its quanta there,  $D_i \div T$ , times its rotation over the rest rotation of the matter quantum,  $\omega_r/\omega_m$  (the energy sources, as a photon weighs  $h\nu/c^2$ ); for the giving, the body's record's two levels at its shell times the bound charge it holds there; for the recoil, the click's tally.  $w$  is the one weight with which the writer's family reads the written family (whoever reads with  $w$  sources with  $w$ : action and reaction one key), and  $E$  the written row's divisor. The five instances are the hold, the source, the giving, the start (the rest of the hold's line, its fixed point) and the recoil, and no number is the act's own:

generic (the rows' keys), vector (one carried division), local (the Node). Under the divisor 1 the level is the count itself, written once and never added interval after interval: the count is read once (Section 7).

**The hold.** A body of quanta  $M_k$  of the family  $k$  has the count  $s = \sum_k w_k M_k$ , the charge  $Q$ , the momentum  $\mathbf{n} = \mathbf{j} \text{ div } T$  with  $\mathbf{j}$  the count's line's current through its Ports (a reading of its record; its velocity  $\mathbf{n}/M$  in Links per interval), the spin  $\mathbf{S}$  the curl of that current about its centre and the moment the same with the charge's sign. At every Node of its support it writes into the families it holds: gravity's time part the count declared there, over the row's divisor  $E_s$ , with a remainder of its own at that Node, never the whole  $s$  at every Node, so that two bodies' wells add and the field's rest is Poisson's with the sources; its vector part  $j_a \text{ div } (E_s T)$  and its tensor part  $(n_a n_b) \text{ div } (E_s M)$ ; the charge's time part the count times the body's sign  $q$  (a family carries no sign; a body's key signs its count, so one matter family holds bodies of both signs) and its vector the same current with the sign; and the dipoles at the Node's six neighbours. The three parts are one source, the count and its current's tensor at one scale with no factor declared: the linearised field's 4 and 2 were nature's numbers by hand and left. At a body with a record the count written is its own record's quanta there,  $D_i \text{ div } T$  with the form's remainder carried, so the well moves with the record to the form's resolution and the self-force vanishes.

**The source.** A record writes as a body does: at its Nodes the sourced family's level gains  $w \times ((D_i \text{ div } T) \times (\omega_r/\omega_m) + r_i) \text{ div } E_s$  each interval, with the same  $w$  the record's family reads the sourced family by; so light sources gravity and the bound charge as its quanta do ( $E = mc^2$  on the comparison side), and every pair of families that read each other is reciprocal. The field's shape is the static limit of Rule3,  $(\Delta - \kappa^2) a = -\sigma/q$  with  $\kappa^2 = 6 \text{ den/num} - 6$ ,  $\kappa$  the inverse reach,  $\sigma$  the source and  $q$  the weight.

**The giving.** A body that gives writes the given family's level at its outer shell, the Nodes with a Port to a Node outside the body: each interval the shell gains  $(M_{\text{pol}} \times a_{\text{body}}) \text{ div } E_s$  at both levels, the family's write by the body's record with the bound charge  $M_{\text{pol}}$  it holds there, so a neutral atom shines, the flux grows as  $M_{\text{pol}}^2$ , and the given row's divisor is the one coupling of light to charge, as  $e$  in nature. The window's three acts are the open (the given family's count at the body down by one, the count's line's move with the opposite sign), the write, and the close at an outward flux of  $T$ , the universe's quantum action, at which the record is named with its direction the tally's sign per axis; one quantum is the train the window writes,  $N_q$  periods of the giver's rotation until the outward flux reaches  $T$ . One quantum per window bounds the charge's divisor from below (Section 7).

**The recoil.** At the close the body's record turns by  $\delta k = \sigma_a \times (k_q \text{ div } M)$  per Link along each axis with a tally, the rotation act by the table's triple nearest  $\delta k$ , the angle's remainder carried at the body, the taker and the giver with opposite signs; so the record's envelope carries the phase  $k$  per Link of a moving body and the velocity moves by the quantum's momentum over the body's quanta, a heavy body's recoil small by its quanta. The momentum  $\mathbf{n}$  is read from the record's current and is no level; a taker with no record of its own takes no recoil.

**What is derived and no primitive.** A body has no law of motion of its own. Its record moves by Rule3 alone, and in a pace gradient its packet accelerates at  $a = \Delta \times |d\omega_0/dc| \times (c_+ - c_-)/D$ ,  $\Delta$  the mode's top velocity,  $d\omega_0/dc$  its redshift per level,  $c_+$  and  $c_-$  the read level at its two faces  $D$  Links apart; the count follows the record by the count's line; bodies with the same  $\Delta \times d\omega_0/dc$  fall alike. The reader's pace carries the read family's vector part,  $-(n_b V_b) \text{ div } W$ , so the record's packet turns and drifts in a vector part's gradient by Rule3 alone: the Lorentz force and its gravitational twin with no line of their own. The spin turns with the record by Rule3 alone; the precession of a

spinning body is a reading of its record. Every primitive of the law is one line of its table [1, the primitives], each an act of Rule3 with declared coefficients and nothing else.

## 5 The stable body

### 5.1 What a body is

A body is its count and its record [1, the stable body]. The count  $c_i$  is the family of clicks' level at the body's Nodes, in quanta; it enters Rule3 through the pace alone,  $p_i = \Gamma$  – the held level at the Node, and it never splits: a quantum moves whole, by the count's line. The record (now, before) of the body's family splits by the send and is bound by the count. A body's declaration names its family, its Nodes, its count per Node, its momentum at its two levels and its spin's two levels where it has one, and on a body that gives its stocks and its emitter, and nothing else: no lowered pair on its Nodes, no seed, no stop, no profile, no closed Port, no period.

### 5.2 The well and the reach

At a Node whose held level is  $c$ , Rule3's one-Node rotation rises from  $\cos \omega_0 = \text{num}/\text{den}$  by  $(1 - p^2/\Gamma^2)(\text{den} - \text{num})/\text{den}$ , and its bonds fall from  $\text{num}/(3\text{den})$  to  $\text{num} p_i p_j / (3\text{den} \Gamma^2)$ : the symmetric form of the line with the weights  $1/p_i$ . The depth is bounded by the family's own  $1 - \cos \omega_0$ , so a family whose pair lies close to 1 binds nothing. A body's clock is set by the level at its Nodes and its side, from the rule: a body's mass is its count.

*The reach of a held family is its pair.* A held family's rest solves  $\text{num} \Delta^2 a - 6(\text{den} - \text{num}) a = -3\text{den} \sigma$ ,  $\Delta^2$  the six-neighbour second difference and  $\sigma$  the source once an interval, so outside a body its rest falls by  $e^{-\kappa}$  per Link with

$$\cosh \kappa = \frac{3\text{den}}{\text{num}} - 2, \quad (8)$$

the family's own band at zero rotation (the evanescent  $k$  of Proposition 1 at  $\omega = 0$ ;  $\kappa^2 = 6\text{den}/\text{num} - 6$  to the second order), the same pair that sets the band of its records and de Broglie's wave number for them: the gap is the mass of the records and the inverse reach of the field, one number. The pair [1, 1] has no gap and its rest is Poisson's over the whole GameBoard, the pull between bodies; a pair with  $\text{den} > \text{num}$  has the gap  $2 \cos \omega = 2\text{num}/\text{den}$  at  $k = 0$ , so at [1, 2] with the divisor 1 the level inside a body is its count and falls by  $e^{-\kappa} = 0.127$  per Link outside ( $\cosh \kappa = 4$ ). A body's well is the static field of every family it holds, read at its Nodes by the paces: a body's binding, a mirror's line and a window are the short-reach family's, and the pull between bodies the massless one's, two held rows of the universe's declaration, one pair each.

### 5.3 The generator

The generator builds every world from zero; it declares nothing and holds no float [1, the generator]. (a) Its input is a world in the law's form: the GameBoard, the Node clock, the universe it names and one body by its Nodes with their counts; the iteration runs on the body's Nodes and their surroundings, out to where the mode's tail falls below one unit. (b) The iteration is Rule3's read act with the before-coefficient 0 on the region, then the division act to the amplitude unit  $A$ : the power iteration of the symmetric form's top mode, the bound mode above the band. (c) The profile is an integer vector bounded by  $A$ , so the iteration is a map on a finite set and enters a cycle; the stop is the first repeat, exact, no tolerance and no declared count. (d) The mode's second level is the read

act once more, halved; the two levels are scaled together so that the record's conserved form equals  $cT$ , the body's count times the quantum's action: the body's record carries exactly its quanta's action. (e) The moving body is the resting envelope with the phase  $k$  per Link along the axis of motion, the rotation act on its two levels by the triple of the pair  $(m, j)$ ,  $\cos k = (m^2 - j^2)/(m^2 + j^2)$ , exact;  $k$  is the pair at which the packet's own current along the axis at the norm  $T$  equals the count's line's velocity of the packet, by bisection on  $j$  at the declared  $m$ . (f) The amplitude unit  $A$  is derived, never written: the bound of Section 2.6 solved for the amplitude; the fixed point does not depend on  $A$  beyond its resolution. (g) The field at rest, the start of a world: the held family's own line is iterated to rest first, the count over the divisor added at the body's Nodes each pass, the homogeneous line outside; the map is monotone, so the levels rise to a fixed point, Poisson's rest with the sources within one unit; a world starts from the field at rest, never from zeros, whose transient rings. The start writes every held family at its rest once before the first interval, the level and its remainder together: the fixed point of the division act is a pair, and a remainder 0 by fiat is no rest.

In the rule's own universe the generator is Rule3 itself (Section 7): a body's bound record is what Rule3 makes of its count at its Node in the rule's own integers, run until the record stands, and its period, amplitude, clock pair, tail and wavelength are readings of the standing record and never inputs.

## 5.4 The velocity

Rule3 is translation-invariant, so the record's wave number  $k$  is conserved where no pace gradient stands: the record's centre moves at its group velocity  $v_g(k)$  exactly, bound or spreading. The count follows the record by the count's line: with the record's norm  $cT$  the current across the body's face is  $cT v_g$  per interval, so  $c v_g$  quanta cross per interval and the count's centroid moves at  $v_g$ , exact up to the remainder [1, the velocity]. The well moves with the count one quantum at a time,  $1/c_i$  of a Link, so the record is never kicked by a whole Link. A one-Node body of one quantum binds nothing and is a free record: its velocity is  $v_g(k)$  with no count to follow it. A larger fall of the bound norm, or a slope that drifts, names a missing law; the body's own numbers are never patched.

## 6 The families from the rule

A family is a band of the one rule and not a declaration, and its row holds nothing the rule and the geometry fix [1, the families from the rule]. This section derives the exact bands from the one-Node line, writes every pair over the one denominator, sorts the families by the three tensor ranks of the three axes, and states under their own names the hypotheses on the quarks, the weak interaction and the electron.

### 6.1 The exact bands

**Proposition 5** (the three exact rotations). *The line of one Node at rest,  $a_{\text{next}} = 2 \cos \omega_0 a_{\text{now}} - a_{\text{before}}$ , is exact in integers, leaving no remainder from any integer start, exactly where  $2 \cos \omega_0$  is an integer. The rotations among them are  $2 \cos \omega_0 = 1, 0$  and  $-1$ , the periods 6, 4 and 3, the pairs  $[1, 2]$ ,  $[0, 1]$  and  $[-1, 2]$ ;  $2 \cos \omega_0 = 2$  keeps a level and  $-2$  alternates it, no rotation.*

*Proof.* With integer levels the right side is an integer exactly when  $2 \cos \omega_0 a_{\text{now}}$  is, for every  $a_{\text{now}}$ , that is when  $2 \cos \omega_0 \in \mathbb{Z}$ ;  $|2 \cos \omega_0| \leq 2$  leaves the five values, of which  $\pm 2$  are the degenerate recurrences. The period of  $a_{\text{next}} = a_{\text{now}} - a_{\text{before}}$  is 6, of  $a_{\text{next}} = -a_{\text{before}}$  is 4 and of  $a_{\text{next}} = -a_{\text{now}} - a_{\text{before}}$  is 3.  $\square$

A record on an exact band at rest at the vacuum's pace leaves no remainder in its rest rotation, so its quanta stand exactly at every Node, it gives nothing and never decays alone: stable matter is built on an exact band. Everything else of it clicks as every family's does: its packets (a wave number is no integer rotation), its Nodes in a well and its reads, so an exact band is seen exactly when it is moved or probed and only then. *The exact bands are exact only in the vacuum:* at a Node of pace  $p_0$  the band's rest rotation is, by Proposition 2 at  $k = 0$ ,

$$2 \cos \omega = 2 - 2 \left( \frac{p_0}{\Gamma} \right)^2 \frac{\text{den} - \text{num}}{\text{den}},$$

an integer at  $p_0 = \Gamma$  alone for the three pairs, so a record on an exact band inside a well, its own or another's, leaves a remainder and clicks like every family's; the three phases of the third (Section 6.4) sum not to 0 but to a share of a record's level that grows with the well, 0.015 at a level 30 of 12,000, 0.148 at 300 and 1.2 at the self-binding edge (COMPUTED), and white matter stands only where the well at its Nodes is small against  $\Gamma$ . Under a divisor far above 1 a white body of thousands of quanta is white to  $10^{-5}$ ; so white matter requires a divisor far above 1, and the smallness of the charge's coupling is a condition of the quarks' whiteness before it is a distance of nature from the rule.

## 6.2 The mass is one integer; the wall is one

Every pair is written over the GameBoard's one denominator  $\Gamma$ ,  $\cos \omega_0 = m/\Gamma$ , the declaration holding the numerator  $m$  alone: light  $m = \Gamma$ , the exact band  $m = \Gamma \text{ div } 2$  ( $\Gamma$  even), matter its one integer, 6,667 in the universe of record (the nearest to  $2/3$ , every number written before a run within  $5 \times 10^{-5}$  of itself). The wall is  $w = 6\Gamma^3$  for every family; the pair enters the coefficients as the file writes it,  $m$  over  $\Gamma$  ( $R_a = 2m p_a^2$ , the rest term  $12(\Gamma - m)p_0^2$ ), and a reduction to lowest terms is a reading and never the loader's act, so every family's remainder has the one resolution and the exact bands leave none in any form ( $S + 6R = 6\Gamma^3$  exactly at the vacuum's pace for  $m = \Gamma \text{ div } 2$ ). With the one wall the amplitude unit of the width falls to  $A = 512,409$  in the universe of record (COMPUTED), and  $T$  is set once so that every record at  $cT$  stays under  $A$ .

## 6.3 The final unification

**Proposition 6** (the three reads of the line). *Of a held family, Rule3 reads three kinds of component and no fourth: its time level, into the clock's pace  $p_0$ ; its vector component  $V_a$ , into the Link's angle; and its aa components, into the axis's pace  $p_a$ . These are a scalar, a vector and the diagonal of a symmetric tensor of rank two over the three axes, with 1, 3 and 6 components under the 48 signed permutations; a component of rank three enters no coefficient of the line, so a family of rank three would be read by no family and is no family.*

*Proof.* By Eq. (4) and the transport's angle (Section 2.7), the coefficients  $w$ ,  $R_a$  and  $S$  and the angle  $k$  are functions of  $\Gamma$ , the pair, and the time levels, the  $V_a$  and the aa components read at the Node and through the Port, and of nothing else. The symmetric tensors over three axes of the ranks 0, 1 and 2 have 1, 3 and 6 independent components.  $\square$

So the three of Rule3 is the three axes, and the three reads are the three ranks of what a family writes: the count, the current and the current's tensor (the hold, Section 4), the writer's side of the same three. Every family is one rank and carries two keys, its pair and its divisor.

**Proposition 7** (the gap and the reach). *A family's band reaches  $\cos \omega = 1$  exactly when  $\text{num} = \text{den}$ ; a held family with  $\text{den} > \text{num}$  has the gap  $2 \cos \omega_0 = 2 \text{num}/\text{den}$  and the finite reach  $1/\kappa$  of Eq. (8), and the held family of unbounded reach is  $[1, 1]$  alone, whose rest is Poisson's.*

*Proof.* By Proposition 1 the band's top is num/den, which is 1 exactly when num = den; and  $\cosh \kappa = 3 \text{ den/num} - 2$  equals 1 exactly then, while for den > num it exceeds 1 and the rest falls by  $e^{-\kappa}$  per Link (Section 5.2).  $\square$

**One free number per rank (DERIVED).** ASSUMPTION, named: the fields that every family reads into its pace at every distance, the pull between bodies and the charge's field, are fields of unbounded reach, and a quantized scalar holds nothing. Then a field every family reads is on  $[1, 1]$  (Proposition 7), and its divisor is free, since Rule3 is linear and its form's scale is free: one count writes one level over  $E_s$  at any  $E_s$ , and no line of the rule fixes it; the vector's divisor is the fine-structure constant  $\alpha$  and the tensor's is Newton's  $G$  (Section 8.2). A quantized scalar holds nothing, has no divisor, and its pair is free: the mass  $m$ . An exact band has both fixed, the pair by Proposition 5 and the divisor 1 by the well being the count, so it carries none. Thus each rank carries one free number and no more, the scalar the mass, the vector  $\alpha$ , the tensor  $G$ : the count three is derived from the three reads, the values free as nature's. The crack, named: a second massive scalar family, a second mass at rank 1, the rule does not forbid; the universe of record declares one, and nature's spectrum of masses is the open row under this line.

A family's parts are the tensor rank of what writes it (1 the count;  $[1, 3]$  the count and the current;  $[1, 3, 6]$  the count, the current and the current's tensor); its phase is 2 where it carries quanta (a current, the count's line, the clicks) and 1 where it never clicks, the highest rank, which every family writes and which reads none, and the exact band, whose rotation leaves no remainder: two real fields. Its reads are the graph of the ranks, each family reading the ranks above it, with the one weight 1 (reciprocity), the reader's own twist and by its own count with the sign of the body that holds it; its held writer writes the count with the body's sign and the dipole of the body's spin with the divisor equal to its phase (1 for the real field, 2 for a pair, Dirac's 2). The action of one quantum is  $T$ , the universe's one integer for every family: Rule3 is linear and its form's scale is free, so no line of Rule3 fixes  $T$ ; it is a unit, declared once. So the declarations hold  $\Gamma$ ,  $T$ , the matter pair and the two divisors, and nothing else of physics.

## 6.4 The third and the conversion

**Hypothesis 1** (the third: the quarks and the strong force). A record on  $[-1, 2]$  turns a third per interval; three thirds in the phases 0,  $2\pi/3$  and  $4\pi/3$ , three consecutive states of one record, sum to zero in their linear write at every interval while their quanta add and stand constant (the white body), and the whiteness is read at every Node from its own three levels, the click's local language. When a third moves off its body every Node between the two carries the three levels' sum uncanceled, a record of the third's own count per Node, so the tube between them costs the third's count per Link and its cost is linear in the length: the tension is the count, no number, the strong force's line; when the tube's quanta reach a third's worth the click converts them into a new pair of thirds, the string breaks into mesons and a third is never free. The three colours are the three phases; a third's charge reads a third, the sense of a third is the sign  $q$  of its read, and a meson is two thirds of opposite sign.

**Hypothesis 2** (the conversion: the weak force). A click that takes a quantum of one family and gives it as quanta of others, the rotations adding as the crystal's row has it, is the whole weak interaction: one Node's act, no field and no massive vector, and no family's row declares anything for it.

A pair's numerator may be negative: the third's  $R_a = 2 \text{ num } p_a^2$  is negative,  $S$  is what the line makes it, the band is the mirror of  $[1, 2]$  ( $2 \cos \omega$  inside  $[-1, 1]$ ) at every wave number and

every pace at or below  $\Gamma$ , the edge  $P = 2\Gamma$  by Proposition 3, the rest term  $36p_0^2$ ), and a pair with  $\text{den} > |\text{num}|$  is admitted as a massive kind; nothing of Rule3 changes. The three colours are three interval states: a third's record at rest steps (now, before)  $\rightarrow$  (–now – before, now) with the period 3, so its three phases are three bodies whose two levels are three consecutive states of one record (7, –10, 3 on one Node: each interval's three levels sum to 0 exactly, the quanta  $7^2 + 10^2 + 3^2 - 7(-10) - (-10)3 - 3 \times 7 = 177$  constant), and the generator writes two levels and no new key.

*A third does not bind itself:* its read is negative, so its own well pushes its rest rotation up into the vacuum's band ( $2 \cos \omega = 0.31$  at a count 3,000 of 12,000, inside  $[-1, 1]$ , COMPUTATION) and the record leaks; a record of the third at one Node is a packet of every wave number and disperses in every universe, its uniform mode alone exact. So one family on  $[-1, 2]$  holds no standing body, no white body and no tube, and the white body of Hypothesis 1 stands only with the colours as three families of one pair under one constraint of whiteness, or, the line taken in the rule's own universe: *the colours are the three axes*. The white body is a pixel of  $[2, 3]$  (Section 7); the count's line carries a tally per axis, so the three colours are the three tallies  $\sigma_x, \sigma_y, \sigma_z$ , and whiteness is their balance at rest (an isotropic tail carries no net current through any Port); under a pull the clicks are biased along one axis at the rate  $\omega_b e^{-\kappa d}$  per interval, which falls with the distance and gives no linear potential: the force between two pixels is short-ranged, its range  $1/\kappa$  (2.2 Links at 3,000 of 12,000; the click's reach 18 Links at  $T = 1$ ), the nuclear force between nucleons and no confinement inside a pixel, which has no inside; the smallest bound fraction of a horizon pixel is  $0.2255/(1/2) = 0.451$ , so a third of a pixel never binds and disperses (confinement is the edge, no line of its own) while two thirds bind, and a meson is two fractions above 0.451 of opposite  $q$  within the tail's reach. The charges  $2/3$  and  $1/3$  as numbers stay open (Section 9).

## 6.5 The electron is the lightest band

**Hypothesis 3** (the electron). The lightest massive band the rule holds is  $m = \Gamma - 1$ ,  $\cos \omega_e = 1 - 1/\Gamma$ ,  $\omega_e = \sqrt{2/\Gamma}$ , and the electron is one quantum on it with  $q = -1$ : no pixel (a pixel's ratio of counts is at most 2.22, Section 7), stable by the integer (one quantum does not disperse into fractions and no massive band lies below), its charge the proton's exactly since  $q$  is the click's integer; the positron is the same band with  $q = +1$ , and pair creation is the conversion.

The mass is the count: the rule reads inertia as the count, the momentum  $n = Mv$  in the unit  $Q$  with the wall  $W = 3QM$ , so  $m_p/m_e = c_p/1$ , the proton a pixel of 1,836 quanta, bound and within the horizon when  $0.2255\Gamma \leq 1,836 \leq \Gamma \text{ div } 2$ , so  $\Gamma$  lies between 3,672 (the horizon pixel) and 8,142 (the edge pixel), and the world's one number is measured; with the stable pixel being one (Section 7) the proton is the horizon pixel,  $m_p/m_e = \Gamma \text{ div } 2$  and the atomic level's  $\Gamma$  is  $3,672 = 6 \times 612$ , one number and no range, a derivation from the law and no prediction of 1,836, which is its input. What this carries by name: one quantum is one energy and its rotation is its frequency, nature's  $E = h\nu$  being the record's click rate; the spin  $1/2$  is open.

## 7 The rule's own universe

**Hypothesis 4** (the rule's own universe). A universe whose every number is the rule's  $[1, \text{the rule's own universe}]$ . Its pairs are the integers 1, 2 and 3 of the three exact rotations and the three axes:  $[1, 1]$  the vacuum's band, on which the two real fields stand;  $[1, 2]$  the exact band;  $[2, 3]$  the matter pair,  $\cos \omega_0 = 2/3$ , the first pair past the exact bands, which the universe of record approximates by 6,667 over  $10^4$ ; and  $[-1, 2]$  the third. Its divisors are 1: the well is the count, every held level is

the count itself, so a count near  $\Gamma \text{ div } 2$  is a horizon, the pace 0, and no body's Node exceeds it, the maximal coupling.  $\Gamma$  is a multiple of 6 so that every pair is exact over it (12,000:  $m = 12,000$ , 6,000, 8,000 and  $-6,000$ );  $T$  is the one unit and cancels in every reading.

So it carries zero free numbers, no  $G$ , no  $\alpha$  and no  $m$ , against the universe of record's three, and it is the same rule on another declaration, a universe of its own beside the universe of record and not in its place. Two named assumptions carry it: the pairs from 1, 2 and 3 (the denominator the three axes), which gives the matter pair  $[2, 3]$  and the crack  $[1, 3]$ ; and the divisors 1, which gives everything below.

## 7.1 The bound body is one Node

Under the divisor 1 a body that binds itself at rest is one Node, a *pixel*, its count  $c$  in  $[0.2255 \Gamma, \Gamma \text{ div } 2)$ : the lower edge is where the one-Node well first holds a rotation below the vacuum's band, the three-dimensional Green's function of the six Ports, a pure number of the rule (below it the record disperses); the upper is the horizon (the pace 0, the Node reads nothing and freezes). Its record is the one-Node line at its bound rotation  $\omega_b(c)$  (0.841 at the edge, 0.464 at 5,412 of 12,000), and its tail outside is the well, evanescent by

$$\kappa = \text{acosh}\left(\frac{9}{2} \cos \omega_b - 2\right) \text{ per Link}, \quad t = e^{-\kappa}, \quad (9)$$

the evanescent wave number of the pair  $[2, 3]$  at the rotation  $\omega_b$  (Proposition 1): 0.45 at 3,000 and 1.27 at 5,000 of 12,000, so the tail ends within three Links (COMPUTED). Its mode carries the count through the form,  $D = \text{now}^2 - \text{next} \times \text{before} = b^2 (2 - 2 \cos \omega_b)$  at rest with both levels  $b$ , so a pixel of  $c$  quanta loads with  $b = \text{isqrt}(c T \text{ den div } (2 \text{ den} - a))$  for the clock pair  $[a, \text{den}]$  of  $2 \cos \omega_b$  (69 at 3,000, 97 at 4,000, 139 at 5,000 of 12,000 at  $T = 1$ ), and its tail as  $b t^{|dx|+|dy|+|dz|}$  rounded ( $t = 0.640, 0.362$  and  $0.281$  at those counts). A body of more quanta than  $\Gamma \text{ div } 2$  is a cluster of such Nodes, each a bound body, clicking to one another.

So between bound bodies there is nothing but the click and the band  $[1, 1]$ : the record does not leave its Node ( $\kappa > 0$ ), a quantum crosses a Link whole at a click and nothing else does, the overlap of two tails within reach is read by the count's line and clicks (the exchange), and light and the two real fields carry the rest; a Node is a bound body or vacuum. The bound body keeps its record on the GameBoard and Rule3 steps it at its Node and along its tail as it steps every record; the one-Node line is the reading of its rotation, not a step of its own. In the universe of record no body binds itself (the well is  $M \text{ div } E_s$ , the edge  $M \geq 0.2255 \Gamma E_s$ ), so its bodies are declared modes and the theorem does not reach them.

## 7.2 The universe is bound; the click joins and parts

**Assumption 1** (the universe is bound). Every body is bound, no body is free, and a record that is in no bound body is a message between two clicks, the one that gave it and the one that will take it: a matter record below the edge cannot be a body and disperses; light never binds, its band's bottom is 0.

A fall is the bias of a bound body's clicks, its tail longer toward the well (the local band lower there,  $\kappa$  smaller), so the remainder crosses  $T$  first on that side. The assumption removes no number; it names the universe that has none, and where it fails (the universe of record, nature) the distances of Section 9 stay.

**The click joins and parts.** Two clusters whose tails overlap (within  $1/\kappa$ , three Links) read each other by the count's line and click, the tail biased toward the deeper ( $\omega_b$  falls with  $c$ ), so the smaller gives quantum by quantum to the larger until its count falls under the edge and it dissolves into messages the larger takes: the join is a run of clicks whose last ends the smaller's record, one Node where  $c_1 + c_2 < \Gamma \text{ div } 2$  and a cluster of two at the horizon otherwise, one record for the joined cluster. A cluster parts where a Node's count falls under the edge or where two tails cease to overlap: the tail's e-fold is  $1/\kappa$ , about two Links at 3,000 of 12,000, but the click's reach is where the Link's current  $a_1 a_2 e^{-\kappa d}$  falls under  $T$ ,  $d = \ln(a_1 a_2 / T) / \kappa$ , about 18 Links at  $T = 1$  for two pixels of 3,000, and the transfer between two pixels runs at about  $\omega_b e^{-\kappa d}$  per interval (0.33 at two Links, 0.09 at five); beyond the reach no click, parted. A taker at the horizon overflows into a neighbour inside its own well, so the cluster grows a Node per click, and a body never splits into three equal pixels ( $2 \times 0.2255 < 1/2 < 3 \times 0.2255$ ). The one record opening for the joined cluster and closing everywhere for the parted one is the half quantum's one non-local act, and there is no other; the conversion changes the family.

**The observer is a cluster.** A Node reads six Ports and a cluster's record is what its Nodes carry, so inside a cluster everything is seen (a GameBoard reading is the local language of its own record) and beyond its tail only a whole quantum arrives, a click.

### 7.3 The algebra of clusters

One closed derivation from the root on the one assumption “only clusters” [1, the algebra of clusters].

- (1) *The Node holds its count; the count is the record's form.* A Node holds one integer, its count  $c$ , and nothing else; in a universe of clusters the count is no level beside the record,  $c = D \text{ div } T$  at the Node with the remainder  $D \bmod T$ ,  $D$  the record's form there, the click is that integer division (the pixelation), and no hold of a declared count stands beside the record; a count below zero is the count parted from its record, a defect of a computation and never a reading.
- (2) *The count is read once.* A count enters a family's pace once, through the held content of the family that holds it at the divisor 1, read down the graph of the ranks at the weight 1, so every family's pace at a Node is  $p_0 = \Gamma - c$  and on its Link  $p_a = p_0 - c - c_a$ , and a pixel's horizon is  $\Gamma \text{ div } 2$ .
- (3) *The edge.* A pixel binds itself where the GameBoard's Green's function of its own well closes, at  $0.2255 \Gamma$  (2,706 of 12,000) on the law, and at  $0.3444 \Gamma$  (4,133) on a line where the level enters the Link once; below the edge a pixel is a cloud and disperses.
- (4) *The tail.* The bound rotation  $\omega_b$ ,  $\kappa$  and  $t$  of Eq. (9), the level  $b t^d$  and the well  $c t^{2d}$  at  $d$  Links; on the law at 3,000 of 12,000:  $\omega_b = 0.8105$ ,  $t = 0.640$ , the wells 1,229, 503 and 206 at one, two and three Links; at 4,000: 0.6578, 0.362, 524, 69, 9; at 5,000: 0.5137, 0.281, 395, 31, 2 (COMPUTED); two pixels bind together while each count with its partner's well at their distance stays within the horizon.
- (5) *The transfer* between clusters is  $\omega_b e^{-\kappa d}$  per interval at  $d$  Links, and the click's reach  $\ln(a_1 a_2 / T) / \kappa$ .
- (6) *The light* is written by clicks alone: the giver's rotation over  $E_s$  at its Nodes, one quantum per window; the charge's divisor is the bound of one quantum per window,  $E_s(\text{charge}) = 0.86 (\Gamma \text{ div } 2)^{3/2}$ , derived from  $\Gamma$  and no free number ( $4.0 \times 10^5$  at  $\Gamma = 12,000$ ); the giver's count is at most  $(A^2 / T)^{1/3}$ .
- (7) *The one integer.* The universe of clusters carries  $\Gamma$  and nothing else, the pixel's size in quanta;  $\alpha / G$  goes as  $\Gamma^{3/2}$  through the charge's divisor.

**The couplings are at the edge.** Every coupling is the edge the rule allows and no number: a pixel's count is at most  $(A^2/T)^{1/3} = 4,447$  and its well at least  $0.2255 \Gamma = 2,706$ , so  $E_s(\text{gravity}) \leq 1.64$  and is 1, and  $E_s(\text{charge})$  is the bound of one quantum per window: a giver's window closes when the outward flux reaches  $T$ , the flux of the written wave is  $a_{\text{light}}^2 \sin \omega \sin k$  per outer Link per interval ( $0.745 a_{\text{light}}^2$  at  $\lambda = 4$ ) and the window is one interval at least, so a given record carries more quanta than the stock loses unless  $0.745 a_{\text{light}}^2 \leq T$ , that is  $E_s \geq g M a_{\text{body}} \sqrt{0.745/T}$  with the write  $g q M a_{\text{body}} \text{div } E_s$ . *The stable pixel is one:* the small gives to the large until it melts, so the bound universe has one stable count, the horizon pixel, and its second body, the electron, is the open distance  $m_p/m_e$ .

#### 7.4 The screen is a cluster; the wavelength is the giver's

Everything that stands on the GameBoard of a look is a bound body or a cluster of them, the source, the screen, the slit's walls, the mirror, the polariser and the detector alike, each pixel at or above the edge; a face of the GameBoard is no body, a click on a face is no click, and a look with a declared element that is no bound body is not a look in the rule's universe. A solid wall of adjacent bound pixels does not exist, two adjacent pixels at the edge pass the horizon on their Link in every case, so a screen is a sieve of pixels at a period  $s$  with the Links between them under the horizon: on the law 4,000 at  $s = 2$  and 3,000 at  $s = 3$  (COMPUTED); a sieve at  $s = 2$  reflects the light of  $\lambda = 4$  (Bragg's condition,  $2s = \lambda$ ) and is the wall and the mirror of the rule's universe, and a row of taker pixels at  $s = 2$  is its screen.

*The wavelength is the giver's.* The given light carries the giver's bound rotation, and on the band  $[1, 1] \cos \omega = (2 + \cos k)/3$  along an axis, so

$$\cos k = 3 \cos \omega_b - 2, \quad \lambda = 2\pi/k : \quad (10)$$

a pixel at the edge ( $\omega_b = \arccos(2/3)$ ) gives  $k = \pi/2$  and  $\lambda = 4$  exactly, the law's wavelength derived; 4.18 at 3,000 and 5.29 at 4,000 on the law; a deeper pixel gives a longer wave, and a giver's declared wavelength is this number rounded to the Link.

#### 7.5 The hierarchy is recursive

A cluster of clusters is the same structure one level up, Nodes each a bound body clicking at one another, with no number that bounds the depth. *The click is the pointer:* the click is the only thing that passes between a cluster and its neighbour and between a level and the next. *The levels at a Node are the nesting:* the three held levels of a Node, the matter's tail, the charge's well and gravity's well, are the three kinds of binding by rank, and the size of a level is the depth in its kind, gravity's well at a Node being the sum of the tails of every body above it (each  $c_i t_i^{2d_i}$ ), so a GameBoard reading names the clusters a Node belongs to; a level  $n + 1$  is bound by the same inequality as the level 1 with the cluster's total count over its members' spacing in place of a pixel's count over one Link, and the one bound on the depth is the horizon at a Node, the content read there under  $\Gamma$ , at which a cluster's centre is the horizon pixel. *Three kinds of binding, not three levels:* quanta into a pixel by the matter's tail (the step in count the pixel's  $c$ , in ticks its period), pixels into a white cluster by the charge's well (the step  $E_s$  over  $q M$ , 37 at  $\Gamma = 3,672$ , the same number as the coupling's gap to  $\alpha$ ), and white clusters into a gravitational cluster by gravity's well ( $e^{\kappa d}$  for tail-touching pixels, the sum's well beyond); the third kind nests without bound, so the count of gravitational levels is the universe's history and no number of the law.

**The ladder of the bound states.** The bound states of a Node at a given  $\Gamma$  are one rung per integer amplitude  $b$  from the edge to the horizon, each carrying the count  $c = b^2 (2 - 2 \cos \omega_b(c))$

at its self-consistent rotation: at  $\Gamma = 24$  there are two rungs ( $b = 5$  and  $b = 8$ ) and at  $\Gamma = 6,000$  about 67 on the law ( $b = 46$  to 112,  $c = 1,396$  to 2,700, the period 7.5 to 13.4 intervals), the ratio of neighbouring rungs  $1 + 2/b$  and the whole ladder spanning 2.22 (COMPUTED), a near continuum that singles out no ratio: neither 1,836 nor the hadrons' 1.31 is predicted, the mass ratio to the electron fixing  $\Gamma$  and nothing more.

**The two readings of  $\Gamma$  are two levels.** Every level has its own  $\Gamma$  and its own interval, a level's interval being the period of the record one level below it, so the proton to electron ratio reads the atomic level's  $\Gamma$  (3,672 to 8,142) while the universe's age reads the top level's  $\Gamma = 6t$  (below); the tick ratio of one level is the cluster's period over the pixel's,  $e^{\kappa d}$ , 7.6 for two pixels of 2,000 at two Links at  $\Gamma = 6,000$ , and about 66 such levels lie between the atomic  $\Gamma$  and the cosmic one (COMPUTED). The chain from the universe's age to the atomic count is not closed: the step per level is a law's number only if the law fixes the stable cluster, and the count of levels and the law tying a level's  $\Gamma$  to its parent's are open by name.

## 7.6 The Node clock is not constant

**Hypothesis 5** (the Node clock grows).  $\Gamma_{t+1} = \Gamma_t + 6$ , the step the six Ports of a Node and no free number.

The rule's physics depends on the ratios  $c/\Gamma$  and  $m/\Gamma$  alone, so every pair steps with  $\Gamma$  ( $[m_0 \Gamma_t \div \Gamma_0, \Gamma_t]$ , the matter +4 and light and the charge +6 per interval) and the wall,  $A$  and the twist's unit are derived from  $\Gamma_t$  each interval, the declaration holding the Node clock as  $[\Gamma_0, 6]$  and the rule no number. A count is the integer its Node holds and stays, so a pixel that does not gather falls under the rising edge  $0.2255 \Gamma_t$  and dissolves after  $(c/0.2255 - \Gamma_0)/6$  intervals, the small giving to the large being the way a body keeps its  $c/\Gamma$ ; the distinct bound counts between the edge and the horizon number  $0.2745 \Gamma$ , one and two thirds more kinds each interval (7 at 24, 1,647 at 6,000). Light on  $[\Gamma, \Gamma]$  does not drift; the clocks of pixels do, a pixel of a fixed count slowing as  $c/\Gamma$  falls, with the rate

$$H = \frac{6}{\Gamma} \left| \frac{d \ln \omega_b}{d \ln(c/\Gamma)} \right|, \quad (11)$$

$5 \times 10^{-4}$  to  $1.3 \times 10^{-3}$  per interval at  $\Gamma = 6,000$  (COMPUTED), the rule's Hubble rate, and  $\Gamma = 6t$  reads the universe's age in intervals; a look reads it as the takers' clock ratio growing by  $(6/\Gamma)(s_{\text{taker}} - s_{\text{giver}})$  per interval. A GameBoard closed on every axis has no rest of the held field under a net count other than 0, so a bound universe closed on itself carries a mean of the held field that drifts by the net count over the Nodes each interval: the hypothesis that the universe expands, under its own name and no look of today.

## 7.7 The well is the count and no field; the generator is Rule3

The threshold is  $\kappa = 0$ , the bottom of the vacuum's band  $[1, 1]$  at  $k = 0$  and the self-binding edge alike, where a tail is infinite and a static field on  $[1, 1]$  is Laplace's, no decay and no rest on a closed GameBoard; no computation of the law and none of the generator stands there: every bound body sits inside its band at  $\kappa > 0$  and light inside  $[1, 1]$  at  $k > 0$ . So in the rule's own universe a held family has no record of its own that steps and no rest that is solved: its level at a Node is the count laid there and nothing else, every interval; the well at  $d$  Links is the matter record's own count there,  $ct^{2d}$ , laid from its form over its period, and gravity's well at a Node is the sum of those tails. Under a divisor above 1 the held field on  $[1, 1]$  is the universe of record's.

*The generator is Rule3:* a body's bound record is what Rule3 makes of its count at its Node in the rule's own integers, seeded at the Node with the count's well held, run on its GameBoard over many periods until the record stands (its levels repeating over a whole period), and the record itself is the body's declared record; its period, its amplitude, its clock pair, its tail and its wavelength are readings of the standing record and never inputs, and a clock pair, a tail or a twist computed in floating point is a defect and no reading. At  $\Gamma = 24$  Rule3 in integers gives the periods 9.6, 11, 12 and 14 at the counts 8 to 11 on the law (the Green's function: 9.55, 10.8, 12.2, 13.8), and the pixels the integers hold are  $b = 5$  carrying 8 to 10 quanta and  $b = 8$  carrying 12 to 15 (COMPUTED). *The count is the record's form over its period:* an integer record's form  $D$  at a Node moves between 0 and  $2c$  within one period, so the count is  $(\sum_{\text{period}} D) \text{ div } (PT)$ , read once per period on the record's own clock, never the form of one interval; a declared count is a reading checked against the loaded record's form at the gate  $|c - D \text{ div } T| \leq 2 \text{ isqrt}(c) + 1$ , refused by name beyond it; a count moves only whole and never below zero, a Node at 0 gives nothing, and a clamp that sets a negative result to 0 creates quanta and is no rule. *The generator has no sink:* Rule3 is reversible and nothing leaves a GameBoard but a quantum by a click at a face; a fill of 0 beyond an open face is a wall, so the unbound part of a seed never leaves; a seed of one Node carries the mode's share  $1/(1 + 6t^2 + 18t^4 + 38t^6 + 66t^8)$  and the rest is debris spread over the GameBoard's Nodes, so the record is read at the Node over many periods and purified by re-seeding from the record within three Links of the Node at its peak's phase, integers alone. The generator of the rule's universe writes bound bodies alone: a body is its count at its Node, its record the bound state of that count and nothing else, and a world's declaration holds counts, their bound records and the band  $[1, 1]$ , and nothing more.

## 7.8 The tree

The root is the law in one line. From it alone: the exact bands, the colours, the tube's linear cost, the edge  $P$ , entanglement as the click and the observer as a cluster. With the named assumption of the pairs from 1, 2 and 3: the matter pair  $[2, 3]$  and the crack  $[1, 3]$ . With the named assumption of the divisors 1: the bound body as one Node with the edge 0.2255, the horizon, the tail, the cluster, and the click that remakes, joins and parts. The universe being bound is an assumption and the conversion a hypothesis. So the algebra is closed on one root, two named assumptions, one assumption of the universe and one hypothesis, with zero numbers, and it is not closed toward nature by name: the three distances  $\alpha = 1/137.036$ ,  $m_e/m_P$  and  $m_p/m_e$  (the rule's universe has one pixel mass scale, its counts within  $[0.2255, 1/2] \Gamma$ ), the charges  $2/3$  and  $1/3$ , the generations and the band  $[0, 1]$ . That the three distances are the rule's is the hypothesis under its own name, of which two lines stand: a divisor far above 1 is what white matter needs (Section 6.1), and one quantum per window bounds the charge's divisor from below (Section 7.3); in the universe of record the same bound reads 32,300 against its declared 40,000 (COMPUTED).

## 8 What the law says of nature

### 8.1 The postulates

1. The world is Nodes and Events. Space is the GameBoard; an Event is a wall crossed by a remainder at a Node: a level stepped, a count moved, a click; nothing else happens.
2. A Node holds bounded local information: its NodeState, fixed in size, read from itself and its six neighbours; no read beyond them, no list that grows with the world, nothing of the past kept. Every physical update uses its own record and the six causally available neighbours

with fixed work and storage for a fixed set of families; a self-field estimated or subtracted from anything global is forbidden, whatever the size of its final answer.

3. Consistency is local and causal: an Event first changes its own Node and travels Link by Link, one Link per interval, each Node updating on receipt under the same rule. The one exception is the click: a record ends at once, whole, at the detector.
4. The causal speed is one Link per interval, built in; every other speed is a rational, a count of Links over a count of intervals.
5. Every physical calculation is on bounded integers; there is no float, no root and no draw in the law; a Node keeps only the law's own numbers: each record's two levels and its remainder, the family's pair and the Node clock from the content there.
6. A measurement is a detector's click, an action of the law on the state: the record ends at the detector and the detector's own record changes. Only a click is compared with nature or pinned as an expectation; displays and the host's readings read state and write nothing.
7. The Inside is the GameBoard, where no one measures; the Outside is the detectors and their clicks, the only thing claimed to represent nature.

## 8.2 The constants

The speed of light is one Link per interval. The quantum of action is one click, its energy the quantum's action  $T$ . Newton's constant is not declared: a body of energy count  $s$  makes the level  $c(r) = s \times$  the reference flux/ $E_s/r$  at the distance  $r$  in Links (the hold's row: the count enters the field's line over the row's divisor  $E_s$ ), and the level slows the clock by the potential  $\Phi = -c/\Gamma$ , so

$$G = \frac{\text{the reference flux}}{\Gamma E_s} \text{ per unit of energy count,} \quad (12)$$

with the reference flux a number of the GameBoard's geometry (about 1.4 for a cube of side 3) and the divisor  $E_s$  the energy unit setting the mass of one unit of level [1, the constants]. A world at a smaller  $\Gamma$  has stronger gravity per unit of content; the ratios between rows carry  $G$  once and cancel it. The quantum of distance is the Link, of time the interval; nothing between two Nodes or two intervals is observed. The potential at a body's Nodes changes by  $G$  per unit of energy count and never by less: the quantum of gravity is the click.

**The three free numbers and their bounds.** The rule's own universe (Section 7) carries none of the three; the universe of record carries all three, and its three are these.  $E_s$  of gravity is fixed by no line of the law, as nature's  $G$  is fixed by none (Rule3 is linear and its form's scale is free); its bound is the law's: at every Node of every body the pace stays positive,  $\text{count} \times (1 + K/E_s) < \Gamma$  with  $K$  the hold's kernel summed over the body (about 51 for a body wider than the reach), so  $E_s > K \times \text{count}/(\Gamma - \text{count})$  at the world's heaviest Node, and the reading's resolution bounds it above. The charge's  $E_s$  is free as  $\alpha$  is, bounded below by the same pace at the bound charge's Nodes and by the train (one quantum at least one period). The matter pair is free as the mass is, bounded by the band:  $\text{den} > \text{num}$ , and the reach  $\cosh \kappa = 3 \text{den}/\text{num} - 2$  fitting the body's width.

## 8.3 The forms of nature the line reaches

Each form below is derived from the line before any run and is a prediction of the law, never a result: the reading named with it is what a detector's click would read, and no such reading is made in this paper. The form of nature beside it is on the comparison side and is never an input [1, the rows against nature].  $U = c/\Gamma$  is the potential at the point named, the clock's shift per held level (Proposition 2);  $u$  is light's group velocity at the wave number in question.

- (a) *The redshift* (DERIVED). By Proposition 2 at  $k = 0$  a clock at the held level  $c$  rotates at  $\omega$  with  $\sin(\omega/2) = (1 - U) \sin(\omega_0/2)$ , so two clocks of one family at the levels  $c_1$  and  $c_2$  have their rotations in the ratio  $(1 - U_1)/(1 - U_2)$ ,  $1 - (U_1 - U_2)$  to first order, the deeper clock the slower; the law's reading through the Link's pace is the root of  $(1 - 2U_1 + 2U_1^2)/(1 - 2U_2 + 2U_2^2)$ . Beside it: Einstein's gravitational redshift,  $1 - \Delta\Phi/c^2$ .
- (b) *The bending* (DERIVED). Along a Link light's speed falls by the level twice, once in the clock and once in the Link (Section 2.5), so past a body at the closest distance  $b$  the wavefront turns toward the body by twice the clock's share, and at a receiver  $L$  Links beyond the centroid is shifted by  $4U_b L$ . Beside it: Einstein's  $4GM/(c^2 b)$ , twice Newton's.
- (c) *The delay* (DERIVED). A round trip past the body is delayed by  $4U_b$  times the path's length within the well's reach, the same two factors on the two ways. Beside it: Shapiro's delay.
- (d) *The perihelion* (DERIVED). An orbit at the semi-axis  $a$  and eccentricity  $e$  advances  $6\pi U_p$  per turn,  $U_p$  the potential at  $a(1 - e^2)$ , the same two factors in the orbit's equation. Beside it: Einstein's  $6\pi GM/(c^2 a(1 - e^2))$ .
- (e) *The moving clock* (DERIVED). A body moving at  $v$  carries the phase  $k$  per Link on its bound band  $\omega(k) = \omega_0 + \Delta(1 - \cos k)$ ,  $\Delta$  its packet's top velocity, and its clock at the moving centre is  $\Omega(k) = \omega(k) - k\omega'(k)$ , the phase rate less  $k$  times the group velocity  $v = \omega'(k)$ ; to second order  $\Omega = \omega_0 - \Delta k^2/2$  and  $v = \Delta k$ , so  $\Omega/\omega_0 = \sqrt{1 - v^2/c_b^2}$  with  $c_b^2 = \Delta\omega_0$ , read as the mean of the front and back click periods at two resting detectors. Beside it: Lorentz's factor, at the body's own light speed  $c_b$  and not the lattice's, since no boost is among the 48.
- (f) *The charge* (DERIVED). Like signs a hill, unlike a hollow; a reader of charge  $q$  reads  $+q d$  in its pace and  $-q(\mathbf{n} \cdot \mathbf{A}) \operatorname{div} W$  on its vector part (the induction, Section 4), so like charges moving together repel less. Beside it: Coulomb's force and the magnetic force.
- (g) *Born, Malus, Bell* (THEOREM). The shares of a click are the current's shares (Theorem 6); a polariser's turn of the record's pair gives  $\cos^2$  and  $\sin^2$  of the angle between the pair and its axis; the pair record gives  $\cos 2(a - b)$  and  $2\sqrt{2}$  (Proposition 4). Beside them: Born's rule, Malus's law, Aspect's correlations and Tsirelson's bound.
- (h) *De Broglie's relation* (DERIVED). A giver of rotation  $\omega_b$  radiates a family whose band holds it,  $\operatorname{num}/(3 \operatorname{den}) \leq \cos \omega_b \leq \operatorname{num}/\operatorname{den}$ , with the wave number from  $\cos \omega_b = \frac{\operatorname{num}}{3 \operatorname{den}} (2 + \cos k)$  and the group velocity  $v = \frac{\operatorname{num}}{3 \operatorname{den}} \sin k / \sin \omega_b$ ; to second order  $k = \omega_b v / c_m^2$  with  $c_m^2 = \omega_0 / (3 \tan \omega_0)$ , the family's own light speed. Beside it:  $\lambda = h/p$  with the rest rotation as the mass.
- (i) *Interference* (DERIVED). A record split at two openings  $d$  apart meets on a screen  $L$  beyond with the two arms' phases  $\mathbf{k} \cdot \mathbf{r}$ : the shares  $\cos^2(\pi dy/(\lambda L))$  under one opening's envelope to first order, the bright strips  $\lambda L/d$  apart, and the lattice's dispersion beyond the first order (COMPUTED: at  $\lambda = 4$  on  $d = 16$  and  $L = 97$  the second bright pair moves from 6 to 8.5 strips of 4 Nodes); two arms rejoined at a splitter of share  $s$  give  $4s(1 - s) \cos^2(\Delta/2)$  at the cross exit,  $\Delta$  the arms' phase difference. Beside them: Young's and Mach-Zehnder's fringes.
- (j) *The fall* (DERIVED). In a pace gradient a bound packet accelerates at  $a = \Delta |d\omega_0/dc| (c_+ - c_-)/D$  toward the well (Section 4), so bodies with the same  $\Delta \times d\omega_0/dc$  fall alike, and across the bound bodies of one family that product differs by the band's own shape, a tenth (Section 9). Beside it: Galileo's fall and the equivalence principle.
- (k) *The pull and the orbit* (DERIVED). The held field on  $[1, 1]$  rests at Poisson's with the sources and at Laplace's outside a body,  $c(r)$  falling as  $1/r$ , so the acceleration falls as  $1/r^2$  and a circular orbit's  $v^2$  as  $1/r$ . Beside it: Newton's pull and Kepler's fall-off; a flat  $v(r)$  would name a missing law.
- (l) *The medium* (DERIVED). Inside a body of pace  $p$  light of wave number  $k$  is slowed to  $v_{\text{in}} = (p/\Gamma)^2 \sin k_{\text{in}} / (3 \sin \omega)$  with  $1 - \cos k_{\text{in}} = (\Gamma/p)^2 (1 - \cos k)$  (Eq. (5)), the index  $u/v_{\text{in}}$ ,

and is reflected whole where  $p < \Gamma \sin(k/2)$ ; through a body moving at  $w$  the wave number is matched at the moving face,  $\omega - kw = \omega_{\text{in}} - k_{\text{in}}w$ , and the speed inside is the band's group velocity there. Beside them: the refractive index and the mirror; Fresnel's drag  $w(1 - v_{\text{in}}^2/u^2)$  stands beside the law's number and is not derived from it.

- (m) *Doppler and Sagnac* (DERIVED). A giver receding at  $v$  from a mirror at rest reads its returned clicks every  $P(u + v)/(u - v)$  intervals for a giving every  $P$ ; a giver moving at  $v$  along a ring of  $N$  Links reads its two arms  $2Nv/(u^2 - v^2)$  apart,  $4A\omega/u^2$  for the ring's area  $A$  and angular speed  $\omega$  to first order. Beside them: the two-way Doppler ratio and Sagnac's  $4A\omega/c^2$ , with  $u$  in place of  $c$ .
- (n) *A free record keeps its wave number* (THEOREM). Rule3 is translation-invariant, so a free record keeps its wave number and rotation exactly and a giver's period is read unchanged at every distance, the rounding's walk at most one unit of level per Node per interval. Beside it: Hubble's drift, for which the law's own line is the growing Node clock of Hypothesis 5.

## 8.4 How the forms are reached, not assumed

The law holds none of the forms it is compared with. The causal speed is one Link per interval by construction, and light's group velocity  $1/\sqrt{3}$  at long wavelength is Proposition 1's, not a constant declared. Planck's constant is the click's unit  $T$ , and  $E = h\nu$  is the record's click rate against its rotation. Newton's constant is Eq. (12), a divisor and the Node clock; the factors 4 in the bending and the delay and  $6\pi$  in the perihelion arise from the level entering the Link twice and the clock once; Lorentz's factor is the dispersion of the body's own bound band read at its moving centre, with the body's own light speed and not the lattice's; de Broglie's relation is the band's group velocity; Born's rule, Malus's law and Tsirelson's  $2\sqrt{2}$  are the ladder's and the pair record's theorems. Each form stands beside the law's own expectation, and where the two differ the difference is written before any run, as in (i) and (l).

## 9 What is open, and what is not claimed

The law's document keeps its open points by name [1, what is open]:

1. Whether a body at rest jitters when its current's swing reaches  $T/2$ .
2. The self-source's cubic term: not in the law, the line is the squares' sum alone.
3. The twist table's small angles: a triple with  $d$  at most  $10^9$  reaches no angle below  $6.3 \times 10^{-5}$  radians.
4. The equivalence principle holds to the band's own shape:  $\Delta \times 2 \tan(\omega_0/2)$  differs by a tenth across the bound bodies of one table; the fall with two light bodies reads the rest.
5. The two thirds' charges: a third with the sign  $q$  reads a third, and the composite read modulo a turn identifies  $-1/3$  with  $+2/3$ , but no line gives the up third  $2/3$  and the down third  $1/3$ .
6. The three generations: a hypothesis under its own name, the three ranks of the exact band  $[-1, 2]$ ; no line gives a rank a mass.
7. The exact band  $[0, 1]$  of the period 4: a family that never clicks and carries no colour, unnamed in nature here.

Beside them stand the three distances of Section 7.8,  $\alpha$ ,  $m_e/m_P$  and  $m_P/m_e$ , of which the last fixes the atomic level's  $\Gamma$  and is not predicted, and the spin  $1/2$ .

Not claimed: any agreement with nature, since nothing is run here; the Lorentz group, which is no symmetry of the lattice; a controlled phase of order one between two records, which Theorem 2 excludes from the acts alone; the count of gravitational levels, which is the universe's history; and

the reach of the divisor 1 in the universe of record. Every hypothesis of the paper is named as one, and the tree of Section 7.8 says what each claim needs beyond the line.

## Declarations

**Funding.** None.

**Competing interests.** None.

**Ethics approval, consent to participate and consent for publication.** Not applicable; no human or animal participants.

**Data and code availability.** The law’s document, the simulator that computes the rule and the figures’ scripts are archived at Zenodo under the concept DOI 10.5281/zenodo.22738746 [17], under the MIT licence; no run of it stands in this paper.

**Author contributions.** The author laid down every postulate and hypothesis of the model, directed the work and wrote the paper.

**The use of a large language model.** The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under his direction and instructions, for the software, the computations, the mathematical derivations and their checks, the physics of the law’s document and the drafting of the text; the model is not an author; the author reviewed and edited every statement and takes full responsibility for the content. No figure was made by a generative tool; every figure is drawn by a script from the definitions.

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## A Where this model stands among discrete models

Deterministic lattice models of physics are old: the lattice gases of Hardy, de Pazzis and Pomeau [8] and of Frisch, Hasslacher and Pomeau [9] recover hydrodynamics from Boolean collisions; the quantum cellular automata of Bialynicki-Birula [10], Meyer [11] and Arrighi [12] step a unitary state on a lattice. This model differs from both in its state and in its measurement: the state is integer levels with a remainder per Node and no complex amplitude, and the measurement is a division of the law and no projection. It differs from the pilot-wave account [13, 14] in that no particle position is kept beside the wave: the count follows the record's current by the count's line, and a body's position is its count's centroid. It differs from spontaneous-collapse models [7] in that the click's residue is the rule's own remainder and no stochastic term. In the ontological-models language [15, 16] the record is the ontic state and the click its one read; the correlation  $2\sqrt{2}$  is reached, and Popescu and Rohrlich's 4 [6] is not, since the second label's shares are a rotation's squares.